A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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Abstract

We turn a language model into a *cellular automaton over token space* — a ring of token cells resampled in place by the model's own windowed conditional $p_r(x_i \mid x_{i\pm r})$, probed by common-random-number damage spreading — and develop it into a *validated* measurement instrument, on the principle of **validation by reproduction**. On the Domany–Kinzel automaton, where the damage field is provably the automaton itself, an independent prediction matches ours **bit-exactly** — zero mismatching cells, with a nonzero off-line control — a calibration with no error bar attached, which a plausible-looking but wrong implementation cannot pass. The instrument also separates elementary CA rules by whether damage survives (d=3.0) and recovers known Markov matrices. Across Pythia-410m checkpoints it detects a **developmental transition**: the token-space Lyapunov exponent goes from seeds disagreeing about its *sign* to **every one of 48 post-transition runs being positive**, at two lattice sizes, all four members of a pre-registered family surviving Benjamini–Hochberg correction. We then *delimit* it, the result we consider most useful: real autoregressive generation absorbs *no* injected token error (P_{persist}=1.000 on three models), because free generation never resamples a token. These quantities characterise an iterated-resampling construction, not deployed decoding — a boundary we draw rather than leave to the reader.

1 Introduction

Language models are evaluated almost entirely as functions: given a context, how good is the next-token distribution? But a masked or autoregressive conditional also *defines a dynamics* — iterate it in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the model. The phenomena a CA vocabulary names — damage spreading [Bagnoli et al., 1992], light cones [Lieb and Robinson, 1972], Lyapunov exponents, attractor censuses [Hanson and Crutchfield, 1997] — are decades old. What has been missing is not a phenomenon but a *measurement instrument whose readings can be trusted*.

Our organizing principle is therefore **validation by reproduction** (§3): before turning the apparatus on a language model we require it to recover values that are independently known. Damage-spreading implementations are conventionally validated against fitted critical exponents, which a wrong implementation can match for the wrong reasons; our strongest rung is instead *exact*, and a bit-exact identity cannot be passed by a plausible-looking estimator. Only then do we report what the instrument measures on trained LMs (§4) and, equally, where its readings provably stop applying (§5, §6).

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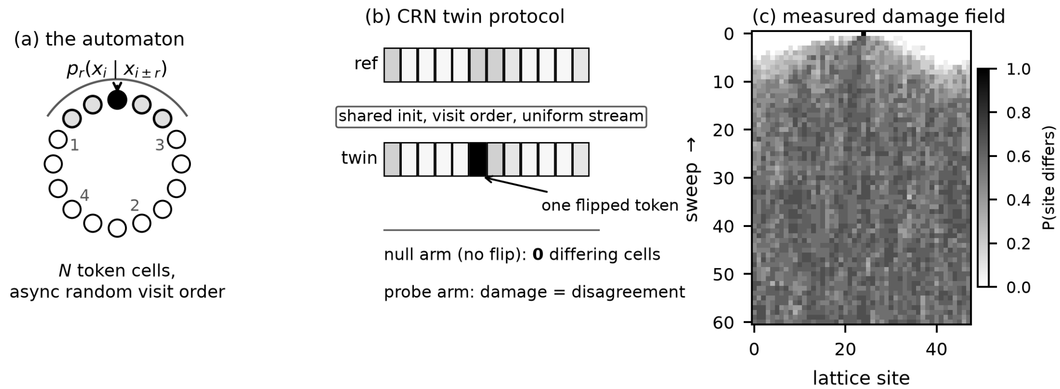

Figure 1: The instrument. (a) A ring of N token cells under async random-order single-site Glauber updates: the model's windowed conditional resamples one cell from its neighbourhood, special tokens forbidden as emissions. (b) The CRN probe: twin lattices share initial state, visit order and uniform stream, so the *null* arm must differ in exactly zero cells — asserted on every back-end, because every damage number depends on it — and the probe arm's site-wise disagreement after one flipped token is the damage field. (c) A measured damage field (T=0.7, r=4, from `results/damage.npz`): damage spreads from the single flipped site into a cone before saturating.

2 The instrument

Figure 1 shows the apparatus. **Rule.** A window of w=$2r$+1 tokens with the centre masked is fed to a masked LM; the tempered centre distribution is the kernel $p_r(x_i \mid x_{i\pm r})$. For an autoregressive model the window is the r cells to the left. **Confounds.** A degenerate low-entropy lattice snaps a perturbation back trivially, so damage D is normalised by the model's own diversity floor D_0 (twins under *independent* noise, no flip), giving D_{norm}=D/D_0. The two necessarily use different couplings — a floor sharing the numerator's is identically zero, since identical twins stay identical — but the cost is bounded: sweeping the shared-draw fraction from 0 to 0.9 moves D_{norm} by 4%. A free control disciplines our language: the temperature "phase transition" is a finite-size *crossover* ($\chi_{\text{peak}} \propto 1/N$).

3 Validation: reproducing known metrics

We check the apparatus against systems where the answer is known, ordered by how much of the instrument's regime each rung shares (Fig. 2). **(1–2)** are smooth-limit unit tests and are labelled as such: a renormalized tangent estimator recovers the logistic map's analytic exponent to machine precision and a coupled-map lattice matches an exact Benettin reference to 1.1×10^{-3}, but neither is evidence about damage spreading, since the twin is re-anchored each step and **a token flip is** $O(1)$.

(3) Elementary CA: a coarse split. The identical protocol on 19 rules of *known* class (12 seeds, rule as the unit) separates damage that *survives* from damage that *dies*. The discriminating quantity is the **ignition probability** — also the directed-percolation order parameter [Grassberger, 1995] — not λ, which averages a bimodal mixture. Ordered rules sit at 0.046 (CI [0.000, 0.102]) against 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest** p=0.0**,** d=3.03. The three-class ordering does *not* survive (p=0.470), so we claim only the coarse split; the demoted group-mean λ ordering, and why it was never a measurement, are in Appendix B.

(4) Domany–Kinzel: an *exact* check. DK [Domany and Kinzel, 1984] is the only rung that is stochastic *and* discrete — the instrument's own regime — and on its p_2=0 line the damage field between two commonly-driven replicas is provably *itself* a DK automaton at the same p_1 [Kohring and Schreckenberg, 1992]. We therefore predict the damage trajectory with an independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \ldots, 1.0\}$ on a ring of 4096 over 1500 steps, **zero mismatching cells**, against 16 in an off-line control at $(0.6, 0.5)$ where the mapping must break. It runs through the same loop as every language-model number below, so indexing, uniform consumption,

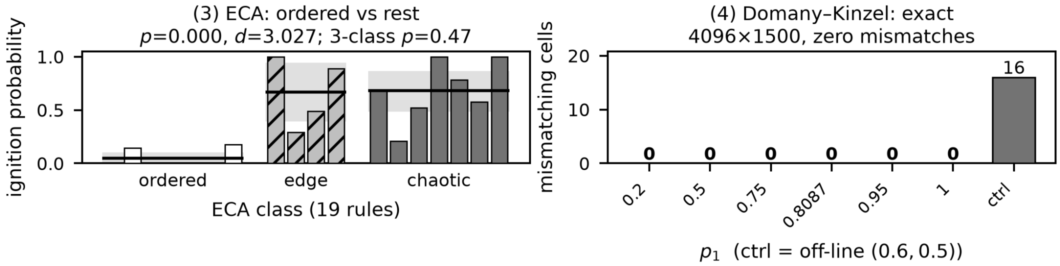

Figure 2: The two rungs that carry the calibration, numbered as in the text. (4) is the load bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, the off-line control beside it showing the test is not vacuous. Rungs (1–2) and (5) are reported in the text.

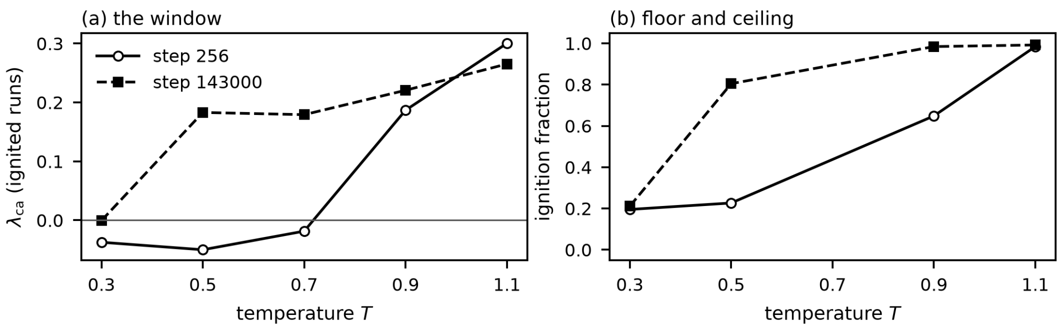

Figure 3: The temperature window (N=48; pre = step 256, post = step 143000). (a) λ_{ca} over ignited runs: at T=0.3 it hugs zero at both ends, while at T=0.9 and 1.1 it is already $+0.19$ and $+0.30$ before the training being measured. (b) Ignition fraction for the swept family: the floor at T=0.3 ($0.20 \rightarrow 0.21$) and the ceiling at T=0.9 and 1.1 (pre ignition 0.65 and 0.98). T=0.7 is the main grid's operating point.

sampling and schedule are verified with no error bar attached; Rule 90 rides inside the same run as a second closed form (0.03125 live cells at t=1500, as obtained). Critical points are a weaker $\sim 1\%$ check: p_c=0.7065 on the site-DP line (published $0.705489(4)$) and 0.8092 on p_2=0, the damage field independently at 0.8089. The published p_2=0 value is disputed ($0.801(2)$ [Zebende and Penna, 1994] vs. $0.8087(5)$ [Hinrichsen et al., 1997]), so we report both and claim neither.

(5) Synthetic Markov sources. The attractor census recovers each model's own sparse transition matrix (row total-variation 0.22) and not the others (cross 0.95; baseline 0.91).

4 A developmental transition in token-space criticality

Pointed at a training run, the calibrated instrument finds a developmental transition: token-space dynamics cross from sub- to super-critical early in training and do not return. Across Pythia-410m checkpoints (`step256`–`step143000`, 8 seeds, N=48 and 96; Fig. 4) the crossing lies between steps 256 and 512 at both lattice sizes, λ_{ca} staying positive thereafter.

The effect occupies a temperature window, and the window has a mechanism (Fig. 3). Repeating the two ends of the curve at $T \in \{0.3, 0.5, 0.9, 1.1\}$ (8 seeds each, BH-FDR over the four) recovers the transition at T=0.5 (p_{BH}=6×10^{-4}) alongside the T=0.7 of the main grid, and at none of $0.3, 0.9, 1.1$ (p_{BH}=0.59, 0.72, 0.59). The ignition fraction makes that a floor and a ceiling rather than a failure: below the window damage barely propagates at either end and λ_{ca} hugs zero, whereas above it the lattice is *already* super-critical before the training being measured, leaving nothing to move; inside the window it swings $0.23 \rightarrow 0.81$ at T=0.5.

Table 1: The pre-registered developmental family. Means over the pre-registered pre and plateau sets; d is Cohen's d; $\lambda_{\rm ca}$ rows exclude unignited runs ($n_{\rm pre}$=15 at N=96). A $p_{\rm BH}$ shown as $<10^{-6}$ rounds to zero at the results file's six-decimal storage.

quantity	N	pre mean	post mean	d	$p_{\rm BH}$
$\lambda_{\rm ca}$	48	+0.0247	+0.1683	1.59	$1.7{\times}10^{-5}$
$\lambda_{\rm ca}$	96	+0.0320	+0.1686	1.71	$2.3{\times}10^{-5}$
$D_{\rm norm}$	48	0.1865	0.5689	2.88	$<10^{-6}$
$D_{\rm norm}$	96	0.1030	0.3062	3.07	$<10^{-6}$

Independent work puts Pythia *generation*'s critical temperature at $T_c{\approx}1$ [Nakaishi et al., 2024]; a different sampler, so a consistent reference point rather than the same measurement — but a ceiling between 0.7 and 0.9 is where it predicts one.

The test was pre-registered (post- vs pre-transition, run-level Mann–Whitney, family = $\{\lambda_{\rm ca}, D_{\rm norm}\} \times \{N{=}48, 96\}$, BH-FDR) with a demotion rule: failure at either size demotes the headline. **All four survive** ($p_{\rm BH} \leq 2 \times 10^{-5}$; Table 1). The pre set is 256/512 and the plateau 2000/8000/143000; one N=96 run is excluded because its damage never ignited, so $\lambda_{\rm ca}$ is undefined for it.

The claim is better stated without any pre/post split, since every effect size depends on where that line is drawn. Before the transition the seeds do not agree on the *sign* of $\lambda_{\rm ca}$ (6/16 negative at N=48, 7/15 at N=96, spanning -0.216 to $+0.320$); after it, **not one of 48 plateau runs is negative**, the minimum being $+0.107$. That is ordinal and uses every run (robustness history in Appendix B).

The curve is not a step, and both ends can be inflated. It overshoots at step 1000; taking the peak as the post value, or the lowest checkpoint alone as the pre value, inflates the effect, so both ends use the pre-registered sets. The overshoot survives correction in one cell of four (accounting in Appendix B), so step 1000 is no distinct phase.

It is not specific to the model we picked. Repeating the grid on four Pythia sizes (70m/160m/410m/1b × 6 checkpoints × 8 seeds, 192 runs) reproduces the transition in **all four** (per-size $p_{\rm BH}$ in Appendix B). Its *timing* moves later with size and then saturates: 70m is already super-critical at the earliest checkpoint probed, 160m crosses between steps 128 and 256, and 410m and 1b both cross between 256 and 512. Learning rate is confounded with size across the suite ($1.0{\times}10^{-3}$, $6.0{\times}10^{-4}$, $3.0{\times}10^{-4}$, $3.0{\times}10^{-4}$ [Biderman et al., 2023]) and the two sizes sharing a rate are the two sharing a bracket, so we report the shift as an ordering, not a size effect. An orthogonal instrument agrees on the onset: Nakaishi et al. [2024] place the emergence of critical structure in Pythia-160m at $k_c{\approx}10^2$ steps by power spectra and part-of-speech correlation — no damage spreading — adjacent to our 128–256 bracket. Both grids are coarse there, so this is agreement to within about a factor of two, not a reproduction. The plateau *level* does not order with size (levels in Appendix B); at four sizes the smallest attainable permutation p is $1/4! = 0.042$, so no capacity axis could have been established, and none is visible. We report that null as the closest thing here to a previously retracted capacity claim.

It is not a restatement of the loss curve. The transition sits where everything else in training changes, so $\lambda_{\rm ca}$ might be an expensive perplexity proxy. It is not: held-out loss falls *monotonically* at all four sizes while $\lambda_{\rm ca}$ overshoots at three, and a non-monotone function of a monotone variable is not a monotone transform of it. They also disagree about *where* (details in Appendix B). We use the curve's *shape* and not its level, the evaluation text not being Pythia's training data.

Which quantity carries the claim. A third lattice size settles it. Over $N{=}48/96/192$ the $\lambda_{\rm ca}$ plateau is **intensive across a** $4\times$ **range** — slope $N^{-0.04}$, varying by 5% of its mean — while $D_{\rm norm}$ falls as $N^{-1.02}$, i.e. $1/N$ (levels in Appendix B). That is what the construction predicts: $\lambda_{\rm ca}$ is a cone-growth *rate* fitted before saturation; $D_{\rm norm}$ is a density ratio whose numerator stays in a cone while its denominator is delocalised. So $\lambda_{\rm ca}$ carries the claim and $D_{\rm norm}$ corroborates at a stated lattice size, never as a lattice-free property. Seed-to-seed variance also collapses $\sim 3\times$ across the transition (Levene $p \leq 1.7 \times 10^{-4}$ at both sizes), an observation rather than a pre-registered test. $\lambda_{\rm ca}$ is continuous and unthresholded, so this sharpening is not an artefact of a discontinuous metric [Schaeffer et al., 2023].

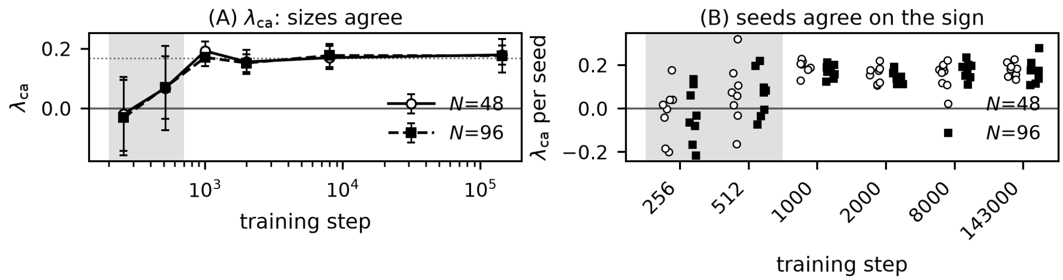

Figure 4: The developmental transition; shading marks the pre-registered pre set. (A) λ_{ca} rises and settles positive, the two lattice sizes on top of each other (levels agree within $\pm 14\%$). (B) Per-seed values: sign disagreement before, consensus after. D_{norm} has the same shape at a size-dependent level.

5 What the damping length measures: free generation absorbs nothing

We tested the instrument's central assumption — that a ring number says something about the model — on *real autoregressive generation*: two CRN twins from a prompt sharing one uniform stream, one forced to a different token at one position, both continuing freely. The null arm diverges by exactly zero, as asserted.

The result is unambiguous and negative. On `pythia-70m`, `-160m` and `-410m` (32 trials each) the injected error **persists in every single trial** (P_{persist}=1.000). Token identity is harsh once an injection shifts alignment, so we also measured distributionally: twin next-token TV, normalised by that between *independent* continuations, settles at $\mathrm{TV}_{\mathrm{norm}} \approx 0.97$.

The mechanism is structural. **Free generation never resamples a token**: once emitted an error stays in context permanently, whereas the ring CA revisits every site, and in-place resampling is what makes healing possible. The damping length, D_{norm} and the light cone therefore characterise *the iterated-resampling construction* and not decoding — which also explains a light-cone velocity depending on r but not the model, and the failure in §6. It does *not* touch §4: the construction is held *fixed* across checkpoints — same ring, radius, temperature, coupling, seeds — so a change measured across checkpoints cannot come from the apparatus. The licence is therefore *comparative*, checkpoint against checkpoint, and never absolute: we do not claim any model *is* critical, since that would need the ring, the radius and D_0 to be principled rather than merely fixed, and they are choices.

Coupling, stated precisely. Damage spreading is a property of *(model, coupling)*, not of the model alone [Kohring and Schreckenberg, 1992, Grassberger, 1995]. Our inverse-CDF sampling is the *monotone* coupling: on DK's binary alphabet it coincides with the maximal-correlation member of the admissible family [Hinrichsen et al., 1997], which is why the identity above is exact; on a vocabulary it does not, so our LM damage numbers sit inside the family rather than at an extremum (measured excess in Appendix B). The choice is justified by replica-independence — each replica's next state depends on its own state and the shared noise alone — which a pairwise maximal coupling lacks. Relative comparisons are unaffected: the coupling is a common mode. **One fact explains both.** Damage-spreading theory was developed on *binary* alphabets — which is why the DK rung is exact, and why its coupling result does not survive $|V| \approx 5 \times 10^4$. The strongest calibration and the sharpest limitation share one cause, so an imported guarantee must be checked against the alphabet.

6 The instrument's boundary: token- vs. activation-space

A black-box instrument earns its keep if its readings track what internals would otherwise be needed to see. Across two model families and two within-model axes they do not: the pairing is family-dependent (Pythia r=+0.71, GPT-2 r=−0.43; the pooled p=0.025 is a Simpson's-paradox artifact), the temperature axis is confounded, and the radius axis is null because $\lambda_{\mathrm{ca}}(r)$ is model-invariant. The reason is structural: the white-box exponent is architectural and *tangent*, while λ_{ca} is *finite*. A clean **negative** — the two levels are distinct and non-proxying.

7 Related work

Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our novelty is the measurement layer, not the framing. Edge-of-chaos measurement is canonical [Langton, 1990, Bertschinger and Natschläger, 2004] and already applied to transformers [Tomihari and Karakida, 2025] and LLMs [Zhang et al., 2025, Li et al., 2025]; we claim neither the concept nor the criticality$\leftrightarrow$capability link, and absorbing-state scaling laws in deep networks are established [Tamai et al., 2025] — we report no universality class. Temperature-driven order–disorder transitions have several prior treatments [Ruan et al., 2026, Sun and Haghighat, 2025, George et al., 2025, Nakaishi et al., 2024, Arnold et al., 2024]; we claim a calibrated route, not the quantity. Perturbation propagation in *activation* space [Petrova and Vejsiu, 2026, Olivieri and Pérez Rodríguez, 2026, Fernando and Guitchounts, 2026] and over compound-AI graphs [Nilayam and Nayak, 2026] is adjacent but in a different space; we avoid "repair" [McGrath et al., 2023, Rushing and Nanda, 2024] and "self-correction" [Petrova and Vejsiu, 2026], categorically different from a spatial damping length. Tang et al. [2026] is the calibration kin; the census descends from Hanson and Crutchfield [1997].

8 Limitations

Neither construction samples the model's actual distribution — masked conditionals are globally inconsistent, the autoregressive rule is resampled in place on a ring — so all claims are properties of the dynamics. Rings are small ($N \leq 384$); the developmental result rests on one model *family* (four sizes and three lattice sizes within it, so a second family is the next requirement) and on a narrow temperature window, outside which the probe saturates either way (§4). D_{norm}'s absolute scale is doubly qualified — it moves with N, and its numerator's coupling is not extremal — so only its relative behaviour is quoted, and census ground truth is synthetic only. **Statistical scope.** The developmental family is pre-registered with BH-FDR at the run level and the ECA test uses the rule as its unit; λ_{ca} statistics exclude runs whose damage never ignited, for which it is undefined, with n stated. The ladder rests on no hypothesis test and the cross-level negative on sign-consistency rather than a surviving p-value. A previously reported capacity$\rightarrow$sensitivity axis was pseudoreplicated and is retracted.

9 Responsible use and conclusion

This instrument measures a construction built *from* a language model, from sampled outputs alone. That is its affordance: it characterises models behind an API, where weights are unavailable and activation-space methods cannot reach. It creates no new generative capability. The misuse we consider most likely is a misreading of our own numbers, which is why §5 is in the body: a damping length measured here describes an in-place-resampling ring, and real generation absorbs *no* injected token error. These quantities must **not** be cited as evidence about the robustness, error tolerance or self-correction of a deployed decoder — a λ_{ca} below zero is not a safety property. That boundary generalises: a metric should be made to reproduce known answers before it is believed about unknown ones. We offer the ladder as a method, not only as our step one — it demoted our own ECA ordering and caught a λ_{ca} reported where it is undefined.

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A Reproducibility

Models. `bert-tiny/mini/base` (L=2–24); `Pythia-14M–1B` [Biderman et al., 2023] and `GPT-2` small–xl; Pythia-410m checkpoints `step256–step143000`. A tiny transformer trained from scratch on tinyshakespeare provides the corpus-checked census. **Grids.** $r \in \{1, 2, 4, 8, 16\}$, $T \in \{0.3, \ldots, 1.5\}$, $N \in \{48, 96, 192, 384\}$, 8 seeds for the developmental family. **Definitions.** $D_{\mathrm{norm}} = D/D_0$ with D_0 the independent-noise no-flip floor; λ_{ca} the log-slope of damaged-site count over the pre-saturation window, with the fit window's constants exposed as parameters and a named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every backend and both update modes through one shared simulation loop; a golden-file regression asserts the core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control that must fail. All code, per-run result files, and figure scripts accompany the submission; every number in the paper is traceable to a result file. **Mirror.** An anonymised snapshot of the tagged tree is at `https://osf.io/9gpu2/?view_only=9973f67d79e7403e96cf4586d6efbdc0`, built from the submission tag by `build_mirror.py`, which rewrites absolute checkout paths and then machine-checks the result for identifying strings.

B Robustness details

Numbers relocated from the body for density; every value below is unchanged from the submitted version and traced to its results file by the number manifest.

Four-size scale grid (§4): the transition reproduces at every size, and the plateau level does not order with size.

size	p_{BH}	plateau λ_{ca}
70m	0.015	0.162
160m	0.003	0.164
410m	2×10^{-6}	0.174
1b	1.5×10^{-5}	0.166

Three lattice sizes (§4): over N=48/96/192 the λ_{ca} plateau is 0.168/0.169/0.160 while D_{norm} is 0.569/0.306/0.139 — the levels behind the body's $N^{-0.04}$ and $N^{-1.02}$ slopes. Seed-to-seed spread at the plateau: per-seed CV 22–25%; it is the checkpoint *means* that agree to a few percent across sizes.

Overshoot accounting (§4): the curve overshoots at step 1000 by 1.4–22.4% across the four cells; taking the lowest checkpoint alone as the pre value would inflate the λ_{ca} effect 1.7×, which is why both ends use the pre-registered sets. The overshoot survives BH correction only in the N=48 D_{norm} cell; on λ_{ca} it is $+1.4\%$ at N=96 (p_{BH}=0.78).

Loss baseline (§4): held-out loss at 26 (model, checkpoint) pairs — the scale grid's 24 plus 410m at steps 8000 and 143000. The steepest loss descent is the 512–1000 bracket for *every* size, whereas the λ_{ca} crossing moves with size and precedes it. Rank correlation is mostly not significant at six checkpoints (ρ from -0.77 to -0.71), so shape and location carry this and not the correlation.

Coupling gap (§5): the excess disagreement of our monotone coupling over a pairwise-maximal one is 1.3–5.4% across the temperature sweep.

Demoted ECA ordering (§3): the group-mean λ ordering we first computed was never a measurement — five of seven ordered rules return exactly $-0.4 \ln 10$, the constant a log-linear fit yields when damage dies immediately.

Estimator history (§4): the ordinal headline survived a later change to unignited-run handling that moved every mean, which is why it, and not any effect size, is the claim.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract states the contribution as a *validated* black-box instrument and names the three things it does: reproduce known metrics before measuring LMs (the load-bearing rung being Domany–Kinzel, where agreement is *bit-exact* rather than fitted), detect a developmental transition across Pythia checkpoints, and *delimit* itself — real generation absorbs no injected token error, so the quantities describe an iterated-resampling construction rather than decoding. The smooth-limit rungs (logistic map, coupled-map lattice) are labelled in the abstract and body as arithmetic unit tests, not validation. The criticality phenomena are framed as decades-old (measured, not discovered) and cited to prior work. The cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed — that earlier two-seed gap is retracted and explicitly not built upon.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work?

   Answer: [Yes]

   Justification: The Limitations section states: neither construction samples the model's actual distribution (masked conditionals are globally inconsistent; the autoregressive rule is consistent but resampled in place on a ring), so all claims are properties of the dynamics; small rings ($N \leq 384$) and one model family for the developmental result, with a second family named as the obvious next requirement; D_{norm}'s absolute scale doubly qualified (it moves with N, and its numerator's coupling is not extremal), so only its relative behaviour is quoted; synthetic-only census ground truth; the windowing scheme as a first-class apparatus factor; temperature scales not comparable across constructions. The statistical scope is stated separately, including that λ_{ca} statistics exclude runs whose damage never ignited — for which the quantity is undefined — with n stated in each case.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

   Answer: [Yes]

   Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are supplied as anonymised supplementary material; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions?

   Answer: [Yes]

   Justification: The anonymised supplement contains the full codebase with a README giving per-phase reproduction commands and measured M1 runtime tables, pinned dependency versions, and idempotent resumable scripts. Every number in the paper is traceable to a committed result file; the analysis scripts assert their emitted group sizes against their declared pre-registered design. A public repository URL will be added at camera-ready if the paper is accepted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details?

   Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps}, \text{seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. **Experiment statistical significance**
   Question: Does the paper report error bars / statistical significance?
   Answer: [Yes]
   Justification: The paper's central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed (p=0.470), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p-value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. **Experiments compute resources**
   Question: Does the paper provide sufficient information on compute resources?
   Answer: [Yes]
   Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. **Code of ethics**
   Question: Does the research conform with the NeurIPS Code of Ethics?
   Answer: [Yes]
   Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. **Broader impacts**
    Question: Does the paper discuss potential positive and negative societal impacts?
    Answer: [Yes]
    Justification: See the Responsible use section. Positive: black-box measurement of trained-LM token dynamics without weight access, applicable to closed/API-only models. It creates no new generative capability. The negative case we treat as most likely is misreading: the paper's damping length and light cone describe an in-place-resampling construction, and real autoregressive generation absorbs no injected token error, so they must not be cited as evidence about deployed-decoder robustness. That boundary is stated in the body, not an appendix.

11. **Safeguards**
    Question: Does the paper describe safeguards for responsible release of high-risk assets?
    Answer: [N/A]
    Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. **Licenses for existing assets**
    Question: Are existing assets (code, data, models) properly credited and licenses respected?
    Answer: [Yes]
    Justification: All assets are publicly released and cited: Pythia and BERT checkpoints (Apache-2.0), GPT-2 (MIT), tinyshakespeare (public-domain source text), WikiText-103 (CC BY-SA 3.0), PyTorch (BSD-3-Clause), HuggingFace Transformers and JAX (Apache-2.0). Usage is research-only and within each licence.

13. **New assets**
    Question: Are new assets introduced in the paper well documented?
    Answer: [Yes]

Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy checkpoints are documented in the repository README and the Reproducibility appendix.

14. **Crowdsourcing and research with human subjects**
    Question: N/A.
    Answer: [N/A]
    Justification: No human subjects or crowdsourcing.

15. **Institutional review board (IRB) approvals**
    Answer: [N/A]
    Justification: No human-subjects research.

16. **Declaration of LLM usage**
    Question: Does the paper describe LLM usage if it was an important, original, or non-standard component?
    Answer: [Yes]
    Justification: Pretrained LLMs are the object of study — they supply the CA update rule — and this is documented throughout. Separately, and disclosed here per venue policy: an AI coding assistant was used substantially in implementing the experiments, in analysis scripting, and in drafting. All experimental designs, pre-registrations and claims were reviewed and verified by the authors against committed result files; several errors introduced during that process were caught by the repository's own assertions and are recorded as retractions in the supplementary findings log.